CAP370:NATURAL LANGUAGE PROCESSING

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: Describe the key concepts, applications, and challenges of Natural Language Processing, including syntax, semantics, phonetics, and morphology.

CO2 :: Apply text preprocessing techniques such as tokenization, stemming, lemmatization, stop-word removal, and text cleaning using Python libraries (NLTK, TextBlob, Pandas).

CO3 :: Analyze sentence structures and semantic meaning and implement word representation methods (BoW, TF-IDF) and parsing techniques (CFG, constituency, dependency parsing)

CO4 :: Demonstrate supervised and deep learning models (RNN, LSTM, Transformers) using TensorFlow/Keras for tasks such as sentiment analysis, text classification, and information retrieval.

Unit I

Introduction to Natural Language Processing : Definition, applications and use cases of NLP, Steps in NLP, Finding structure of Words, sentences, syntax, and semantics, Phonetics, morphology, pragmatics, phonology, Natural Language processing with NLTK

Unit II

Language Models and Text Processing : Text Preprocessing with Python: Reading & Writing files, string manipulation, Regular Expression, Text Cleaning, Tokenization using NLTK, Text Analysis Libraries of Text Processing, Native Python, Textblob, Pandas, Stop-word removal, stemming, and lemmatization., Word Representations: Bag of Words (BoW), TF-IDF using sklearn

Unit III

Syntax and Parsing Techniques : Grammar Formalisms: Context-Free Grammar (CFG), Chomsky Normal Form, Parsing Techniques: Implementing parsers in Python using libraries like nltk, Applications of Parsing: Sentence structure analysis, Semantic role labeling

Unit IV

Semantic Analysis : Lexical Semantics in Python, Discourse Semantics in Python, Semantic Similarity: Ontology and Knowledge Representation, Semantic Web libraries in Python (e.g., RDFLib), Implementing sentiment analysis using Python libraries (TextBlob, VADER)

Unit V

Advanced NLP Techniques : Machine Learning in NLP: Supervised and unsupervised methods using sklearn, Deep Learning in NLP: Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) using TensorFlow/Keras, Transformers and Transformers library, Pre-trained Models: Fine-tuning BERT, GPT System Overview, Text Generation: Basics of Natural Language Generation (NLG) with Python

Unit VI

Prompt Engineering : Core Principles, Types of prompts, Comman Challenges & Solutions, Advance Techniques: Zero-Shot Learning, Few-Shot Learning, COT (Change of Thought), Reinforcement through iteration

List of Practicals / Experiments:

Practical Applications of NLP

- Write a Python script to demonstrate basic NLP operations using NLTK, including tokenization, POS tagging, and chunking.
- . Implement a script to identify phonetic representations and morphological structures of words using NLTK or related libraries.
- Create a script to perform the following tasks on a dataset:Read and write text files. Clean text (remove special characters, numbers, etc.).Tokenize sentences and words.
- Use Python's re module to perform pattern matching on text, such as extracting email addresses, phone numbers, and hashtags.
- Compare text preprocessing using NLTK, TextBlob, and Pandas. Perform stop-word removal, stemming, and lemmatization on a given dataset.

- Implement Bag of Words (BoW) and TF-IDF vectorization using sklearn. Use the vectors to compute cosine similarity between sentences.
- . Define a Context-Free Grammar (CFG) for a small subset of English and use NLTK to parse sentences according to the grammar.
- Write a Python script to perform constituency and dependency parsing on sentences using spaCy
- Develop a Python script to analyze stock prices.
- Create GUI for weather forecast
- Perform sentiment analysis on a dataset (e.g., movie reviews) using TextBlob and VADER. Compare the results of both libraries.
- Use Python's rdflib library to create and query a simple ontology. Demonstrate basic RDF and SPARQL operations.
- Implement a Recurrent Neural Network (RNN) or LSTM to classify text sequences (e.g., classify movie reviews as positive or negative) using TensorFlow/Keras.
- Use Transformers to fine-tune a BERT or GPT model for a text classification task (e.g., spam email detection).

Text Books:

1. MACHINE LEARNING by TOM M. MITCHELL, Mc Graw Hill Education

References:

1. UNDERSTANDING NATURAL LANGUAGE PROCESSING by T V GEETHA, PEARSON